

VUNISEA, Fiji

WMO: 916930

Lat: 19.050S

Lon: 178.167E

Elev: 33

StdP: 100.93

Time Zone: 12.00 (PCF)

Period: 86-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.9	18.9	12.9	9.3	21.3	13.9	10.0	21.7	11.1	23.6	10.2	23.5	0.4	120	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.1	30.9	26.4	30.4	26.1	30.0	25.9	27.2	29.9	26.7	29.5	26.4	29.1	3.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	29.1	26.0	21.4	28.8	25.6	20.9	28.4	86.2	30.2	84.4	29.7	82.7	29.3	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.2	8.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.1	26.7	27.2	27.0	26.4	24.8	23.9	23.1	23.0	23.5	24.1	25.4	26.4
	DBStd	1.91	0.90	0.91	0.95	1.05	1.12	1.27	1.18	1.36	1.61	1.25	1.15	1.07
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5509	518	481	527	491	457	417	405	404	406	437	461	507
	CDD18.3	2468	260	248	268	241	199	167	147	146	156	178	211	248
	CDH23.3	17329	2442	2487	2591	2060	1122	687	424	449	659	821	1446	2141
CDH26.7	3071	533	666	629	374	93	40	18	19	50	60	215	376	
Wind	WSAvg	3.4	3.3	2.8	2.8	3.4	3.6	3.6	3.2	3.1	3.7	3.9	3.3	4.1
Precipitation	PrecAvg	2113	286	244	268	230	159	117	94	95	114	140	159	208
	PrecMax	2951	661	425	417	410	360	290	162	247	219	264	406	393
	PrecMin	1425	55	63	123	68	46	37	24	24	19	35	52	71
	PrecStd	385	141	91	81	99	81	63	38	53	55	61	80	72
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.6	31.5	31.1	29.3	28.7	28.2	28.2	29.5	29.2	30.4	31.0
		MCWB	26.3	26.7	26.7	26.5	25.1	24.7	24.0	23.8	24.8	24.6	25.4	26.2
	2%	DB	30.3	30.8	30.7	30.1	28.4	27.8	27.0	27.2	28.0	28.1	29.5	30.1
		MCWB	26.0	26.4	26.3	26.0	24.5	24.4	23.3	23.4	24.2	24.0	25.1	26.0
	5%	DB	29.7	30.3	30.1	29.5	27.7	26.9	26.1	26.4	27.1	27.3	28.7	29.3
		MCWB	25.7	26.2	26.1	25.5	24.0	23.8	22.4	22.9	23.6	23.4	24.7	25.4
	10%	DB	29.2	29.7	29.5	28.7	27.0	26.1	25.3	25.6	26.1	26.6	27.9	28.5
		MCWB	25.3	25.9	25.8	25.1	23.6	23.2	21.9	22.3	22.9	22.9	24.2	24.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.8	27.7	27.4	26.1	25.6	25.2	24.7	25.4	25.8	26.5	27.2
		MCDB	29.9	30.3	30.2	30.1	28.2	27.7	26.8	27.3	28.1	27.9	29.2	29.8
	2%	WB	26.7	27.1	27.0	26.6	25.5	24.8	24.1	24.1	24.7	24.9	25.7	26.5
		MCDB	29.4	29.9	29.7	29.3	27.6	27.1	26.1	26.3	27.3	27.2	28.4	29.1
	5%	WB	26.2	26.7	26.6	26.1	24.9	24.3	23.5	23.5	24.1	24.2	25.2	26.0
		MCDB	28.9	29.5	29.2	28.6	27.0	26.4	25.5	25.7	26.5	26.3	27.8	28.6
	10%	WB	25.8	26.3	26.2	25.7	24.4	23.8	22.9	22.9	23.6	23.6	24.7	25.5
		MCDB	28.4	29.0	28.8	28.1	26.4	25.7	24.8	25.1	25.8	25.8	27.1	28.0
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.1	4.9	4.4	4.3	4.1	4.7	4.7	4.6	4.7	4.7	4.3
		MCDBR	5.7	5.9	5.9	5.7	5.5	5.4	5.9	5.8	5.5	5.7	6.0	5.3
		MCWBR	2.8	2.8	2.9	2.8	2.9	3.0	3.1	3.0	3.1	3.1	3.0	2.8
	5% WB	MCDBR	5.3	5.5	5.4	5.2	4.7	4.9	5.1	5.0	5.1	5.1	5.4	4.8
		MCWBR	2.9	2.9	3.0	2.9	2.8	3.0	3.4	3.2	3.1	3.2	3.1	2.7
Clear-Sky Solar Irradiance	taub	0.381	0.383	0.379	0.375	0.369	0.369	0.359	0.359	0.373	0.378	0.376	0.379	
	taud	2.639	2.638	2.663	2.662	2.635	2.623	2.634	2.643	2.590	2.591	2.623	2.632	
	Ebn at Noon	963	951	932	898	866	848	868	901	923	944	961	965	
	Edh at Noon	101	99	94	88	84	82	83	88	99	103	102	101	
All-Sky Solar Radiation	RadAvg	6.44	6.16	5.53	4.62	4.04	3.56	3.80	4.32	5.13	5.70	6.14	6.37	
	RadStd	0.56	0.41	0.34	0.35	0.28	0.27	0.21	0.40	0.47	0.43	0.57	0.34	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.18	N/A	N/A	N/A	N/A	+0.20	+0.24	N/A	N/A	+67	+66

Nomenclature: See separate page